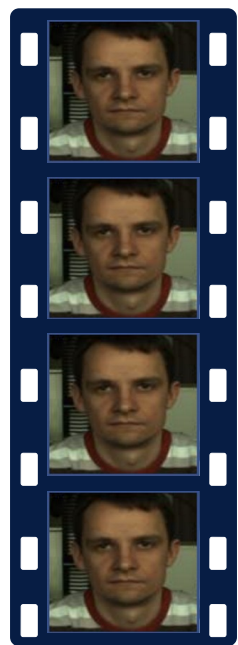


1 Input Video

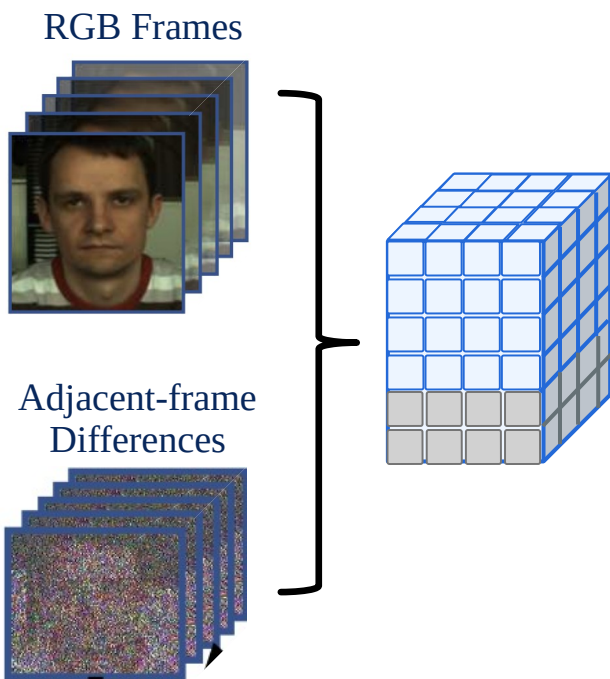


Face Detection
& BBox Expansion
& Crop and Resize

160 Frames
(128 × 128)

cropped facial sequence

2 Fusion Stem



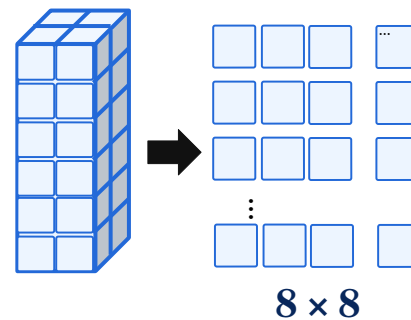
5×5 Conv + BN + ReLU + Pool

3×3 Conv + BN + ReLU

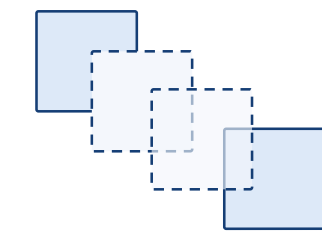
Temporal difference
features

3 Patch Embedding

Frame-wise 2D Conv



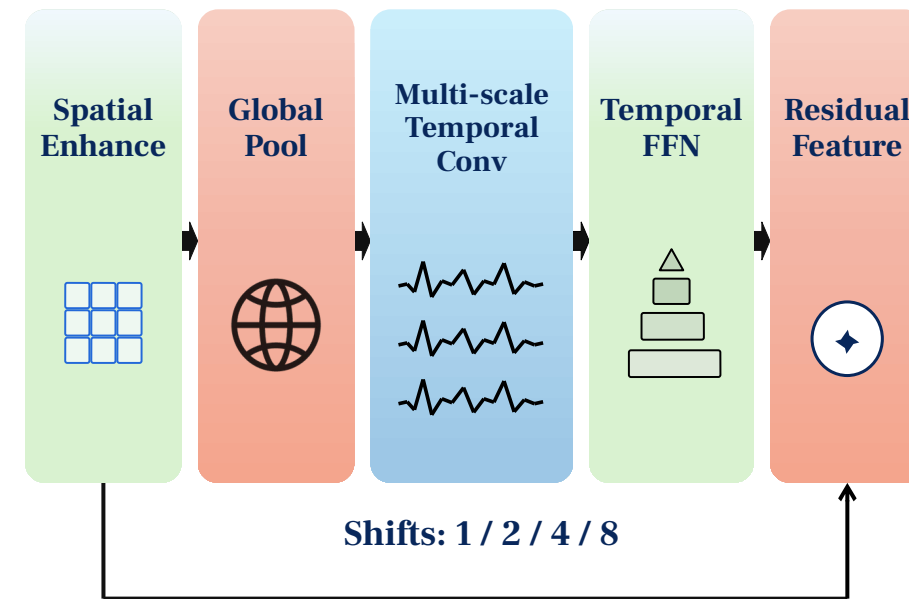
Overlapping Patches



Output: $T \times 8 \times 8$

No explicit positional
embedding

4 Hierarchical Temporal Modeling



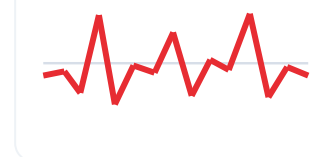
TPT block extracts and fuses Multi-scale
Temporal Features

Increasing temporal granularity suppresses noise

TPT block with FFN module

5 rPPG Predictor

MLP
Projection



BVP / rPPG
Waveform
(160 × 1)

Model output

6 HR Post-processing

BVP Detrend

First-order Butterworth
0.75–2.5 Hz

Welch Power
Spectral Density



Heart Rate (bpm)

Inference only